SleepSense: Machine Learning-Based Prediction of Non-Restorative Sleep in Middle-Aged and Older Adults Using Lifestyle and Health Data

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***Abstract*—SleepSense is a proposed research prototype for predicting a later self-reported sleep outcome from routine health and lifestyle measurements. The study uses an archived dataset from 49,483 residents of Fukushima Prefecture who attended checkups in 2014 and 2015. The team's completed exploratory analysis selected 13 baseline predictors and established a stratified 80/20 split. The minority code of the 2015 outcome accounts for 12.26% of records. The planned study will compare Logistic Regression, Random Forest, and XGBoost under common stratified evaluation conditions. Preprocessing, feature selection, and synthetic oversampling will be confined to training folds. Evaluation will report precision, recall, F1, ROC-AUC, and PR-AUC, with bootstrap uncertainty. SHAP explanations and conventional feature rankings will be presented in a research dashboard alongside a baseline-input demonstration. The target is a single-item self-report, and the proposed system is intended for research demonstration. Model training, performance evaluation, and dashboard development are planned.**

***Index Terms*—non-restorative sleep, supervised learning, class imbalance, data leakage, model explainability, SHAP**

I. Introduction

Non-restorative sleep (NRS) is the subjective experience of waking unrefreshed. It represents one dimension of sleep quality and has been investigated in relation to physical and mental health in middle-aged adults [1], [2]. SleepSense examines whether routinely collected baseline data can support out-of-sample prediction of a later self-reported sleep outcome, how modelling choices affect that prediction, and which inputs contribute to it.

A. Background of the Study and Related Work

Lin *et al.* studied NRS in relation to later type 2 diabetes in a cohort of 3,665 male workers, and Morimoto *et al.* studied its association with self-rated physical and mental health in a postal survey of 33,902 adults [1], [2]. These studies motivate the research question by using NRS as an exposure. Other studies use NRS as an outcome. Wakasugi *et al.* examined combined lifestyle factors, Hidaka *et al.* studied exercise and activity across age and gender groups, and the Nagahama Study considered a broader range of clinical and lifestyle factors [3], [4], [5].

The SleepSense dataset comes from Hidaka *et al.*, who examined lifestyle patterns across two checkups and their association with poor subjective sleep quality in 2015 [6], [7]. That analysis included lifestyle changes across both years. SleepSense instead restricts the feature set to data available in 2014 and evaluates prediction on held-out records. The source article describes the outcome as subjective sleep quality, measured by a single question about getting enough sleep. The repository uses NRS variable names. This proposal uses NRS as shorthand for the archived outcome and retains neutral numeric codes where the 2015 coding is uncertain.

In related machine-learning work, Huang and Huang used NHANES data, tree ensembles, and SHAP for a sleep-disorder outcome [8]. That study informs the proposed explanation workflow, while its target and population differ from SleepSense. Table I summarizes the reviewed evidence. The literature scan is preliminary. Its scope supports a focused benchmark for this dataset.

B. Problem Statements

RP1: Predictive performance. The selected dataset has association analyses, while the performance of supervised prediction from 2014 variables alone remains to be established for SleepSense.

RP2: Imbalance and model selection. Only 12.26% of records carry the minority outcome code. Overall accuracy can conceal poor detection of this class. The effects of preprocessing, class weighting, oversampling, feature selection, and model family require a controlled comparison. Baseline-only feature selection and fitting inside the training partition are essential to prevent leakage [9].

RP3: Interpretation. Model scores alone give little insight into which inputs drive predictions. SleepSense will compare SHAP attributions, built-in importance, and linear-model coefficients. The intended audience includes sleep and public-health researchers, data scientists, and course evaluators.

C. Research Questions

1) **RQ1.** How accurately can supervised learning models predict subsequent NRS from baseline lifestyle and health data?
2) **RQ2.** How do preprocessing, feature selection, and algorithm choice affect classification of the imbalanced target?
3) **RQ3.** Which baseline variables contribute most to model predictions according to explainability methods?

TABLE I
COMPARISON OF RELATED STUDIES AND THE SLEEPSENSE DESIGN

Study	Design and data	Contribution or limitation	SleepSense design choice
Lin *et al.* [1]	3,665 male workers, median follow-up 14.6 years	Uses NRS as an exposure for diabetes incidence	Predict the later sleep outcome in the selected cohort
Morimoto *et al.* [2]	Postal survey of 33,902 adults aged 40–64	Relates NRS to self-rated health in cross-sectional data	Use an earlier baseline and evaluate individual predictions
Wakasugi *et al.* [3]	Combined lifestyle score in 243,767 checkup participants	Supports lifestyle as a predictor domain	Retain individual baseline variables for interpretation
Hidaka *et al.* [4]	Activity and exercise study in 90,122 residents aged 40–74	Reports associations by age and gender	Compare algorithms on a shared holdout and report by gender code
Matsumoto *et al.* [5]	Nagahama Study of 9,788 community residents	Identifies several clinical and lifestyle correlates	Evaluate available baseline predictors and acknowledge omitted variables
Hidaka *et al.* [6], [7]	49,483 Fukushima residents with 2014 and 2015 checkups	Examines lifestyle changes and 2015 subjective sleep quality	Restrict input to 2014 and reserve a test set for final evaluation
Huang and Huang [8]	NHANES, ensemble models and SHAP	Demonstrates explanation methods for a related sleep outcome	Compare linear, bagged, and boosted models on the archived NRS target

TABLE II
ALIGNMENT AND MEASURABLE COMPLETION CRITERIA

Problem, question, objective	Planned evidence	MVP output	Deadline
RP1, RQ1, RO1	Held-out metrics with bootstrap intervals and comparison to majority-class and logistic baselines	Model metrics, confusion matrices, and ROC and precision–recall curves	Week 10
RP2, RQ2, RO2	Controlled factor comparisons with common splits and seeds; uncertainty on differences	Experiment summary and ablation plots	Week 12
RP3, RQ3, RO3	SHAP summaries, dependence plots, local explanations, and comparison with conventional rankings	Global and local explanation views	Week 13

D. Research Objectives

General objective. Develop and evaluate reproducible supervised models that predict the 2015 sleep outcome from 2014 baseline data, and interpret the inputs contributing to those predictions.

Specific objectives.

1) **RO1.** Train and evaluate models with precision, recall, F1, ROC-AUC, and PR-AUC, and quantify uncertainty against a majority-class baseline.
2) **RO2.** Compare Logistic Regression, Random Forest, and XGBoost under a common split and evaluation protocol, and measure the effects of imbalance handling, preprocessing, and feature selection.
3) **RO3.** Identify influential baseline variables using SHAP and feature-importance methods, and compare ranking stability across folds or seeds.

Table II links each objective to its evidence, dashboard output, and planned deadline.

II. METHODOLOGY

A. Study Scope and Current Status

The project covers secondary analysis of the archived CSV, a reproducible training and evaluation pipeline, and a research dashboard. The scope is limited to tabular baseline data and internal validation in the Fukushima sample. Clinical use, causal inference, new data collection, external validation, deep learning, sensor inputs, and mobile deployment are outside the study scope. The exploratory audit, feature allowlist, and stratified split are complete according to the team's source proposal. Model selection, training, final evaluation, explanations, and dashboard implementation are planned.

B. Development Process and System Architecture

The team plans seven two-week Agile sprints with a shared board and milestone reviews. Zuhra Rahimova leads machine learning and data analysis, Fotima Rahimova leads literature review and research, and Aziza Hamidova leads planning and documentation. A protected Git branch requires peer review. Python, pandas, NumPy, scikit-learn, imbalanced-learn, XGBoost, SHAP, matplotlib, Streamlit or an equivalent framework, and pytest form the proposed toolchain. The team will pin dependencies and seeds and save configurations, splits, model files, metrics, and plots. Unit tests will cover preprocessing, leakage checks will reject 2015 inputs or fitting outside training folds, and a smoke test will exercise the dashboard.

Figure 1. Proposed SleepSense system architecture (planned design)

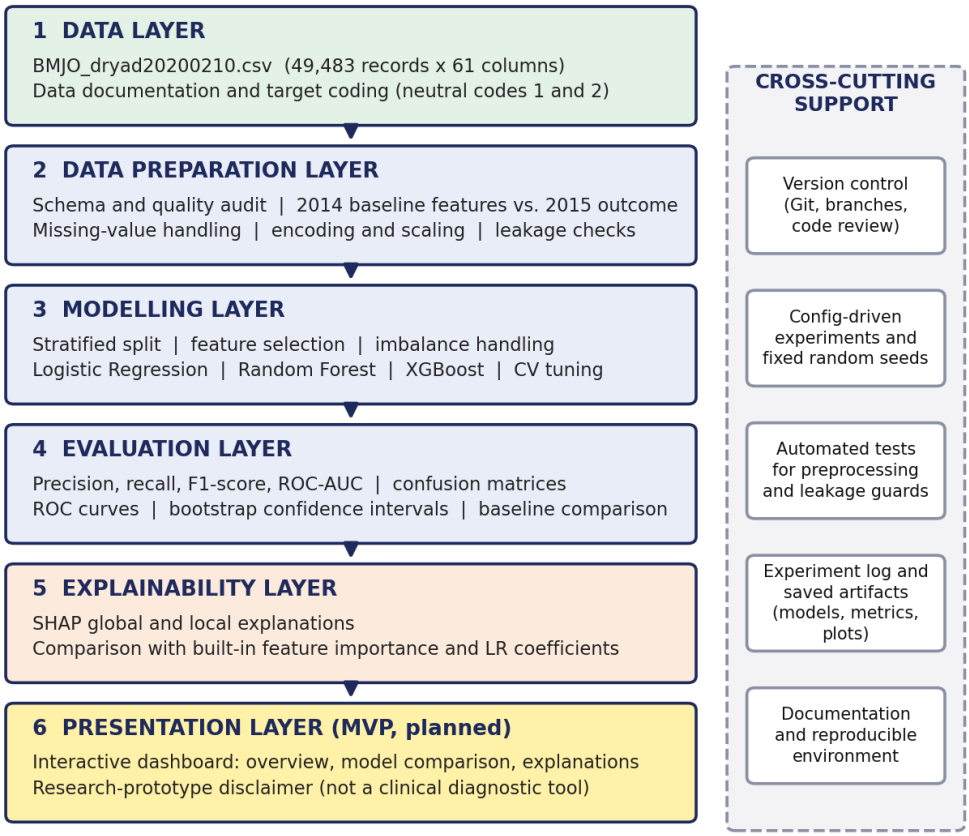


Fig. 1. Original SleepSense system architecture, from data storage through dashboard presentation.

TABLE III
SYSTEM ARCHITECTURE RESPONSIBILITIES

Layer	Responsibility	Output
Data	Version the source CSV and document fields and outcome codes	Raw data and documentation
Data preparation	Apply the baseline allowlist and fixed split, audit leakage, and encode or scale inputs	Feature matrix, target vector, and saved partitions
Modelling	Apply selection and imbalance strategies inside folds; train and tune classifiers	Trained models and tuning logs
Evaluation	Compare models and baselines using locked test data	Metrics, curves, matrices, and intervals
Explainability	Compute SHAP and built-in rankings and assess stability	Attributions and importance tables
Presentation	Load saved artefacts and accept baseline inputs for a research demonstration	Interactive dashboard

Fig. 1 shows the planned six-layer architecture. Table III specifies each layer's output. Version control, testing, documentation, and experiment logging support all layers.

C. Dataset, Outcome Coding, and Baseline Predictors

The dataset is BMJO_dryad20200210.csv, archived in Dryad [7]. It covers residents of Fukushima Prefecture, Japan, with checkups in 2014 and 2015. The team's initial audit is summarized in Table IV. The modelling task uses the target `2015NRS` and the 13 baseline inputs in Table V.

Dryad documentation consulted by the team labels code 1 as absent and code 2 as present for the 2014 NRS variable. The team's source proposal did not independently confirm the corresponding 2015 mapping. The analysis therefore designates code 2 as the positive class by convention and reports

TABLE IV
COMPLETED DATASET AUDIT AND SPLIT

Item	Finding reported by the team
Size	49,483 records and 61 columns
Integrity	No missing values, duplicate rows, or duplicate participant IDs
Age	Source eligibility 40–74 years; observed 2014 age 41–74 years
Gender	Code 1: 26,087; code 2: 23,396
2015 outcome	Code 1: 43,414 (87.74%); code 2: 6,069 (12.26%)
2014 outcome	Constant at code 1; excluded for zero variance
Partition	Stratified 80/20 split, seed 42: 39,586 train and 9,897 test

TABLE V
THIRTEEN BASELINE PREDICTORS GROUPED BY DOMAIN

Domain	Fields and meaning
Demographics	`gender`, `2014Age`
Anthropometry	`2014BMI` (body mass index)
Blood pressure	`2014SBP`, `2014DBP`
Glycaemic markers	`2014FPG`, `2014HbA1c`
Smoking	`2014smoking` (coded)
Activity	`2014EX`, `2014PA` (coded)
Eating habits	`2014meals` (coded)
Alcohol	`2014DF`, `2014AI` (coded)

code-level metrics. Interpretation of code 2 as NRS remains conditional on verifying that mapping. The constant `2014NRS` column is excluded. Questionnaire categories will be used as supplied, with neutral labels where verbal definitions remain unverified.

D. Leakage-Safe Preprocessing and Feature Selection

The fixed split precedes fitting. The allowlist excludes the target, identifiers, all 2015 measurements, and variables that encode changes across years. Each participant appears once in the audit. Pipeline components will be fitted inside training folds and applied to validation and test records without refitting. Range checks will flag implausible values for inspection. Median or most-frequent imputation will be available as a training-fitted safeguard. Logistic Regression will use one-hot encoding for categorical inputs and standardization for continuous inputs. Tree models will use categorical representations appropriate to the implementation. Correlation screening and variance inflation factors will inform interpretation of linear-model coefficients.

The full feature set will be compared with a filter-selected subset (e.g., mutual information) and an embedded subset derived from L1 Logistic Regression or tree importance. Recursive feature elimination is optional. All selection occurs within folds. The imbalance experiment compares unadjusted training, class or positive-class weights, and training-fold-only SMOTE [10]. Validation and test distributions remain unchanged. Thresholds will be chosen using validation predictions and compared with the default 0.5 threshold. Fig. 2 shows the split, leakage controls, and output path.

E. Model Training and Experimental Design

The majority-class dummy classifier establishes a baseline. Logistic Regression supplies an interpretable linear comparator, Random Forest models non-linear interactions with bagged trees [11], and XGBoost supplies a regularized boosting model [12]. Table VI lists the tuning space. Randomized search will optimize ROC-AUC with stratified five-fold cross-validation on the 39,586 training records. All models will use common partitions and seeds.

The comparison will start with default classifiers on all 13 predictors. Next, the team will vary the imbalance strategy and feature set in separate stages, tune the leading configurations, and evaluate a small combined grid. Repeated-seed robustness checks will use development data. Hyperparameters, selection rules, and thresholds will be frozen before the final held-out comparison.

F. Evaluation, Uncertainty, and Explainability

The 9,897 held-out records will be evaluated once after model selection. Metrics are precision, recall, F1, ROC-AUC, and PR-AUC, supported by confusion matrices and ROC and precision–recall curves. For the designated positive code, precision is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$, recall is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, and F1 is the harmonic mean of precision and recall. The precision–recall no-skill reference is the positive-class prevalence. The team will use 95% bootstrap confidence intervals and paired resampling of the same test records to estimate differences between models. Subgroup reporting by gender code will help characterize uneven performance.

SHAP provides local additive feature attributions [13]. For the selected tree model, the study will report mean absolute attributions, beeswarm summaries, dependence plots, and examples for individual held-out records. Logistic Regression coefficients and odds ratios will complement its SHAP analysis. The team will compare SHAP rankings with built-in importances or coefficients and examine stability across folds or seeds. Correlated predictors can share attribution. These explanations describe model behaviour within the study data.

G. Ethics, Data Governance, and Risks

The study uses publicly archived checkup data for academic research. The team will avoid re-identification and will keep dashboard demo inputs transient. The outcome is self-reported, recorded with one item, and drawn from a single prefecture and one year pair. Predictor coverage excludes factors such as stress, depressive symptoms, and sleep duration. External validation is outside this phase. Table VII records the principal risks and planned controls.

H. MVP Dashboard and Delivery Plan

The MVP will load saved artefacts to show the dataset profile, model comparisons, an explanation view, and a prediction demo. The demo will accept the 13 baseline variables and display the model-estimated probability of code 2 with a research-use notice. The about view will explain the data source, protocol, and limitations. Fig. 3 preserves the original

Figure 2. Planned leakage-safe model pipeline and inference flow

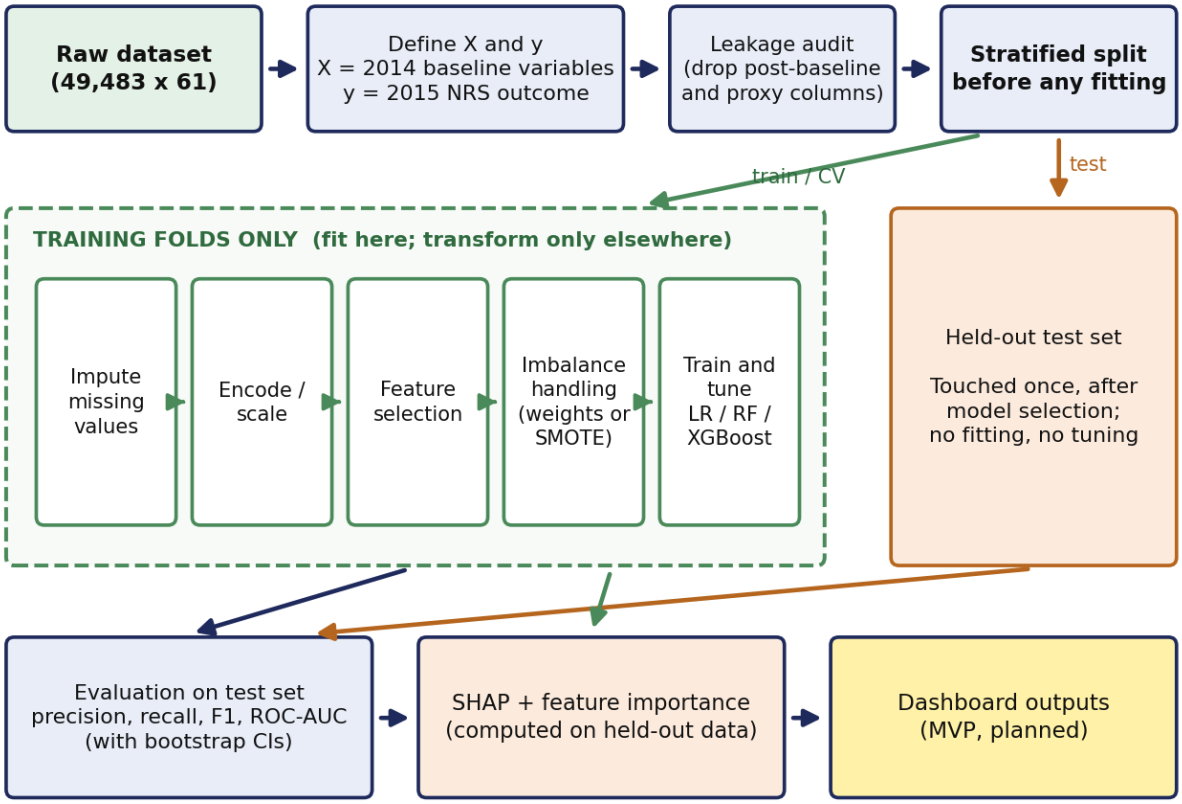


Fig. 2. Original planned model pipeline, with fitting restricted to training folds and a held-out final evaluation.

TABLE VI
MODEL ROLES, TUNING TARGETS, AND EXPERIMENTAL FACTORS

Model or factor	Role	Tuning targets or levels
Logistic Regression	Interpretable linear baseline	L1/L2 penalty, regularization strength, compatible solver, class weighting
Random Forest	Bagged non-linear ensemble	Tree count, depth, leaf size, split features, class weighting
XGBoost	Boosted non-linear ensemble	Tree count, depth, learning rate, row and column subsampling, regularization, positive-class weight
Imbalance handling	Compare minority-class performance	None, class weighting, training-fold SMOTE
Feature set	Measure effect of feature selection	Full 13 predictors, filter subset, embedded subset
Decision threshold	Compare operating points	Default 0.5, validation-tuned threshold

conceptual wireframe. Its chart placeholders describe planned outputs. Table VIII gives priorities.

Fig. 4 shows the original schedule, and Table IX records milestones and owners. The completed Week 1–3 work includes the initial audit, baseline predictor selection, constant-column exclusion, and stratified split. The remaining work is planned.

TABLE VII
PROJECT RISKS AND PLANNED CONTROLS

Risk	Impact	Planned control
Incomplete outcome or questionnaire coding	High	Preserve neutral codes and qualify the interpretation of code 2
Temporal and preprocessing leakage	High	Baseline allowlist, split-before-fit pipelines, fold-local transformations, and leakage tests
Minority-class underperformance	Medium	Class-specific metrics, PR-AUC, weighting and SMOTE comparison, and validation threshold selection
Limited predictors and population coverage	Medium	Report uncertainty, subgroup metrics, and generalizability limits
Tuning or dashboard schedule overrun	Medium	Staged experiment design and weekly reviews against prioritized milestones

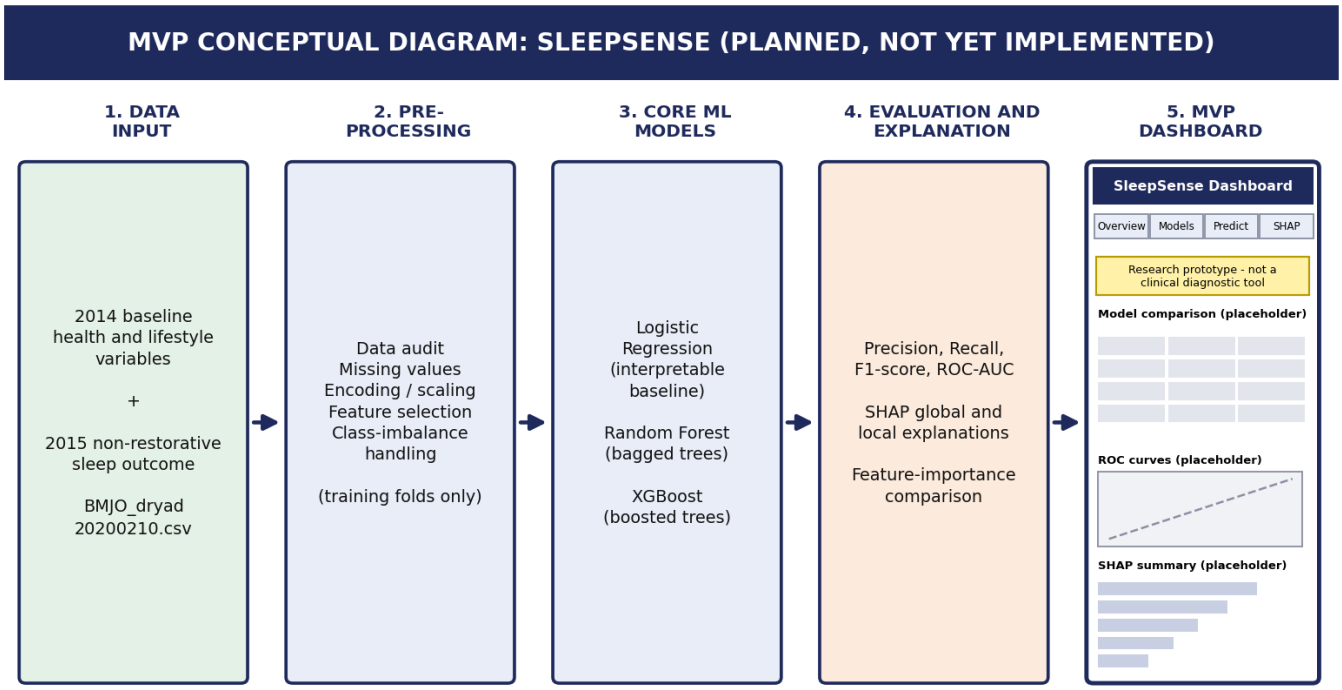


Fig. 3. Original five-stage MVP concept and dashboard wireframe with placeholder plots.

TABLE VIII
PLANNED MVP VIEWS AND PRIORITIES

View or capability	Priority	Planned content
Overview and about	Must	Dataset counts, predictors, split, outcome distribution, protocol, limitations, and research-use notice
Model comparison	Must	Metrics and confidence intervals, confusion matrices, curves, and imbalance and feature comparisons
Prediction demo	Must	Form for all 13 baseline inputs and estimated positive-code probability
Explainability	Must	Global SHAP and feature rankings with model comparison
Local explanations	Should	Explanations for selected test records and demo inputs
Optional extensions	Could	Exportable summary and a threshold slider showing precision–recall trade-offs

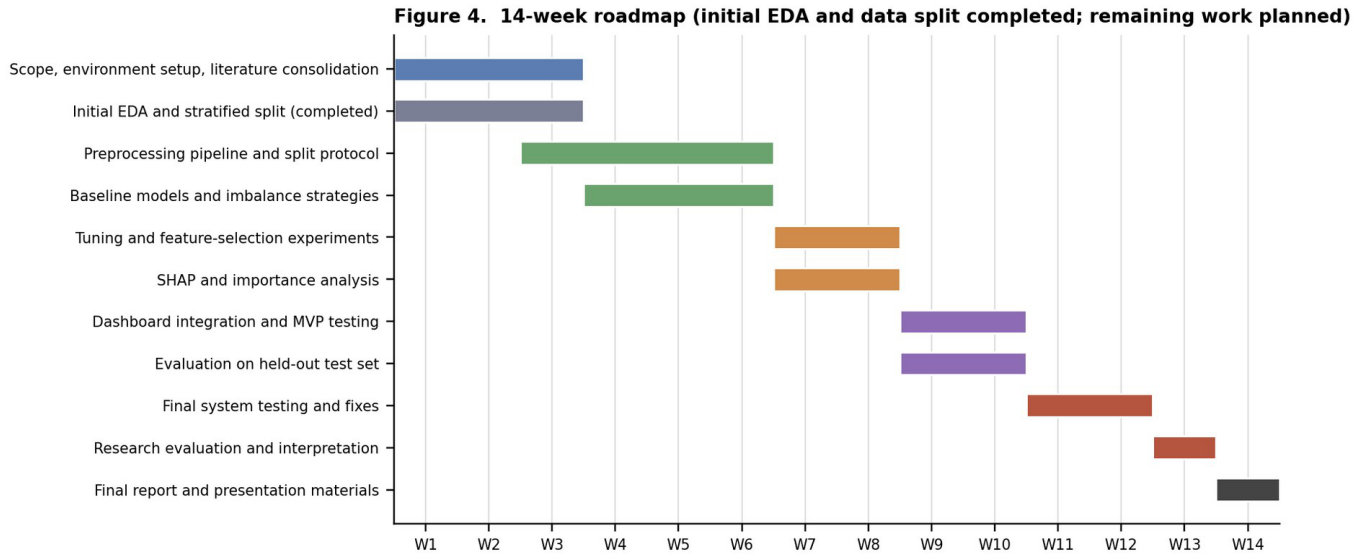


Fig. 4. Original 14-week roadmap. The exploratory audit and split are complete; later stages are planned.

TABLE IX
IMPLEMENTATION MILESTONES AND RESPONSIBILITIES

Weeks	**Milestone**	**Lead or responsibility**
1–3	Initial audit and split completed; finish documentation, leakage checklist, literature consolidation, and environment setup	Zuhra Rahimova (data), Fotima Rahimova (literature), Aziza Hamidova (planning)
4–6	Build preprocessing, tests, dummy and logistic baselines, tree models, and imbalance variants	Zuhra Rahimova; Aziza Hamidova (test records)
7–8	Tune candidate models; implement feature selection, bootstrap metrics, and explanation modules	Zuhra Rahimova; Fotima Rahimova (interpretation notes)
9–10	Integrate dashboard, test the MVP, and run the frozen holdout comparison for RO1	Zuhra Rahimova; Aziza Hamidova (dashboard documentation)
11–12	Finish regression tests and development-seed robustness checks; consolidate RO2 comparisons	All members
13	Consolidate RO3 explanations, sensitivity analyses, and limitations	Fotima Rahimova and Zuhra Rahimova
14	Finalize the report, repository documentation, presentation, and submission package	Aziza Hamidova with all members

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